

# Module 1: Reduced resistive MHD

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The version-1 model is two-dimensional, incompressible, two-field reduced resistive magnetohydrodynamics on a doubly periodic Cartesian domain:

$$(x, y, t) \in [0, 2\pi) \times [0, 2\pi) \times [0, T].$$

$$\mathbf{q}(x, y, t) = (\psi, U), \quad \boldsymbol{\mu} = (\eta, \nu, A_0).$$

- ▶  $\psi$  is the magnetic-flux function and  $U$  is the flow stream function.
- ▶  $\eta$  is magnetic diffusivity,  $\nu$  is kinematic viscosity, and  $A_0$  sets the initial perturbation.
- ▶ The initial trajectories are synthetic reduced-MHD solutions, not genuine M3D-C1 output.

# Two potentials define the physical fields

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## Magnetic field

$$\mathbf{B}_\perp = \nabla\psi \times \hat{\mathbf{z}} = (\psi_y, -\psi_x),$$
$$J = -\nabla^2\psi.$$

$J$  is the out-of-plane current variable.

$$\nabla \cdot \mathbf{B}_\perp = 0, \quad \nabla \cdot \mathbf{v}_\perp = 0.$$

Thus divergence constraints hold identically. The project predicts  $(\psi, U)$  and derives  $(J, \omega)$ .

## Fluid velocity

$$\mathbf{v}_\perp = \hat{\mathbf{z}} \times \nabla U = (-U_y, U_x),$$
$$\omega = \nabla^2 U.$$

$\omega$  is the out-of-plane vorticity.

# The Poisson bracket represents advection

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$$[f, g] = f_x g_y - f_y g_x.$$

The stream-function velocity converts transport into a bracket:

$$\mathbf{v}_\perp \cdot \nabla f = (-U_y)f_x + (U_x)f_y = [U, f], \quad \frac{Df}{Dt} = f_t + [U, f].$$

$$[f, g] = -[g, f], \quad \int_\Omega [f, g] \, dA = 0,$$

Thus nonlinear brackets redistribute fields and exchange energy, but do not directly dissipate it.

# Two PDEs govern the dynamics

$$\psi_t + [U, \psi] = \eta \nabla^2 \psi,$$

$$\omega_t + [U, \omega] = [J, \psi] + \nu \nabla^2 \omega.$$

- ▶  $[U, \psi]$  transports magnetic-flux contours with the fluid.
- ▶  $\eta \nabla^2 \psi = -\eta J$  permits resistive slippage and topology change in current layers.
- ▶  $[U, \omega]$  transports vorticity, while  $[J, \psi]$  is magnetic forcing from the Lorentz force.
- ▶  $\nu \nabla^2 \omega$  smooths vorticity and dissipates kinetic energy.

# Alfvén normalization sets the scales

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$$V_A = \frac{B_0}{\sqrt{\mu_0 \rho_0}}, \quad t_A = \frac{L_0}{V_A}.$$
$$\eta = \frac{\eta_m}{L_0 V_A}, \quad \nu = \frac{\nu_d}{L_0 V_A}.$$

$$S = \frac{1}{\eta} = \frac{t_\eta}{t_A}, \quad \text{Re} = \frac{1}{\nu} = \frac{t_\nu}{t_A}, \quad \text{Pm} = \frac{\nu}{\eta}.$$

$S$  and  $\text{Re}$  compare resistive and viscous diffusion times with  $t_A$ . Neural standardization is a later numerical transformation, not physical nondimensionalization.

# Periodicity and energy close the model

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Periodic faces and PDE-required derivatives match. Gauge and mean-flux conditions are

$$\boxed{\langle U \rangle(t) = 0}, \quad \frac{d}{dt} \langle \psi \rangle = 0.$$

The exact periodic energy law is

$$\boxed{\frac{d}{dt} \frac{1}{2} \int_{\Omega} (|\nabla \psi|^2 + |\nabla U|^2) dA = -\eta \int_{\Omega} J^2 dA - \nu \int_{\Omega} \omega^2 dA \leq 0.}$$

Finite  $\eta$  enables reconnection in current sheets. This is not full M3D-C1 equivalence.

# Module 2: Verified MHD trajectories

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For one parameter vector, the discrete trajectory is

$$\mu = (\eta, \nu, A_0), \quad \mathcal{Q}_\mu^{N_x, N_y, \Delta t} = \{\psi, U, J, \omega\}_{nij}.$$

The solver advances only the prognostic Fourier state

$$\hat{\mathbf{q}}(t) = \begin{bmatrix} \hat{\psi}(t) \\ \hat{\omega}(t) \end{bmatrix}, \quad \hat{U} = -\frac{\hat{\omega}}{k^2} \ (k \neq 0), \quad \hat{J} = k^2 \hat{\psi}.$$

- ▶  $\hat{U}_{00} = \hat{\omega}_{00} = 0$  fixes the gauge and compatibility condition.
- ▶ Reconstructing  $(U, J)$  at every stage prevents the four fields from drifting apart.
- ▶ A run becomes data only after verification, convergence, and case-level acceptance.

# Island coalescence tests every PDE term

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The version-1 periodic benchmark begins from

$$\psi_0 = \sin x \sin y,$$

$$U_0 = A_0(\cos x - \cos y),$$

$$J_0 = 2\psi_0, \quad \omega_0 = -U_0, \quad \mathbf{v}_{\perp,0} = (-A_0 \sin y, -A_0 \sin x).$$

The flow drives two negative-current islands toward the central X-point  $(\pi, \pi)$ :

approach  $\rightarrow$  field compression  $\rightarrow$  thin current sheet

$\rightarrow$  resistive reconnection  $\rightarrow$  merged topology.

**Baseline:**  $\eta = \nu = 10^{-3}$ ,  $A_0 = 0.1$ ,  $N_x = N_y = 128$ ,  $\Delta t = 10^{-3}$ , nominal  $T = 8$ , with post-peak coverage.



# Fourier operators exploit the periodic square

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On the half-open grid  $x_j = 2\pi j/N_x$ ,  $y_\ell = 2\pi \ell/N_y$ , no periodic endpoint is duplicated. For each Fourier mode,

$$\widehat{f}_x = ik_x \widehat{f}, \quad \widehat{f}_y = ik_y \widehat{f}, \quad \widehat{\nabla^2 f} = -k^2 \widehat{f}.$$

## Linear operations

$$\widehat{U} = -\widehat{\omega}/k^2, \quad \widehat{J} = k^2 \widehat{\psi}.$$

They are algebraic in Fourier space.

This Fourier–grid–Fourier cycle is why the method is called *pseudo-spectral*.

## Nonlinear bracket

$$[f, g] = f_x g_y - f_y g_x.$$

Derivatives are transformed to the grid, multiplied pointwise, and transformed back.

# Dealiasing and RK4 define the solver

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Quadratic products can fold unresolved content into false low modes. The frozen operator is

$$\widehat{\mathcal{B}(f, g)} = \mathcal{P}_{2/3} \mathcal{F}(f_x g_y - f_y g_x),$$

where the rectangular mask retains

$$|m| \leq \lfloor N_x/3 \rfloor, \quad |n| \leq \lfloor N_y/3 \rfloor.$$

The semi-discrete state  $\dot{\mathbf{q}} = \mathcal{L}(\mathbf{q})$  uses classical fourth-order Runge–Kutta:

$$\mathbf{q}^{n+1} = \mathbf{q}^n + \frac{\Delta t}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4).$$

Every stage reconstructs  $(U, J)$ , evaluates dealiased brackets, reapplies the mask, and enforces zero modes. Stability monitors do not replace temporal refinement.

# Diagnostics test balance and topology

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The integrated energy defect audits signs and dissipation:

$$\delta_E(t) = \frac{\left| E(t) - E(0) + \int_0^t [\eta \int J^2 dA + \nu \int \omega^2 dA] ds \right|}{\max(E(0), \epsilon_{\text{mach}})}.$$

Topology is measured from tracked O- and X-points:

$$\Delta\psi_{OX} = \psi_X - \psi_O, \quad \Phi_{\text{rec}}(t) = \Delta\psi_{OX}(0) - \Delta\psi_{OX}(t), \quad \mathcal{R}_{\text{rec}} = \frac{d\Phi_{\text{rec}}}{dt}.$$

Resolution evidence also includes  $\|J\|_\infty$ , current-sheet width in grid intervals, spectral-tail energy, symmetry, means, periodicity, divergence, and the definition defects  $J + \nabla^2\psi$  and  $\omega - \nabla^2 U$ .

# Only converged cases become data

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Verification proceeds from exact operators to nonlinear trajectories:

derivatives and Poisson inversion → analytic fields → bracket identities  
→ exact decay → manufactured solution  
→ balances, refinement, and language parity.

- ▶ Temporal refinement recovers fourth-order RK4 behavior before spatial error dominates.
- ▶ Spatial refinement stabilizes fields, peak times, reconnection, sheet width, and spectra.
- ▶ The horizon includes the first resolved peak and a post-peak interval.

Accepted HDF5 cases store fields, coordinates, parameters, diagnostics, configuration, provenance, and individual acceptance flags. A finite run or convincing contour plot is insufficient.

# Module 3: Preserve dataset identity

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Module 3 makes accepted output auditable without changing the reference fields.

## Physical case

`physical_case_id`

Equations, domain, conditions, parameters, and intended horizon.

## Numerical realization

`case_id`

Solver, grid, time step, precision, language, configuration, and revision.

`release`  $\supset$  physical cases  $\supset$  realizations  $\supset$  snapshots  $\supset$  grid points.

Refined, parity, restart, and migrated realizations of one physical case remain linked. This distinction prevents leakage.

# Canonical HDF5 arrays have explicit meaning

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Every accepted case declares

```
schema_name=pinn_m3dc1.case,  schema_version=1.0.0,  axis_order=t,x,y.
```

## Numeric datasets

```
coordinates/{t,x,y}  
fields/{ $\psi$ ,  $U$ ,  $J$ ,  $\omega$ }  
parameters/{ $\eta$ ,  $\nu$ ,  $A_0$ }  
diagnostics/
```

## Scientific metadata

```
physics/  numerics/  
verification/  provenance/  
 $q[n, i, j] = q(t_n, x_i, y_j), \quad q \in \{\psi, U, J, \omega\}.$ 
```

Canonical reference arrays are finite float64 values with shape  $[N_t, N_x, N_y]$ . A non-square, non-symmetric sentinel catches silent Python–MATLAB axis transposes.

# Whole-case splits prevent leakage

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Generalization is measured on disjoint physical-case sets:

$$\mathcal{P}_{\text{all}} = \mathcal{P}_{\text{tr}} \cup \mathcal{P}_{\text{va}} \cup \mathcal{P}_{\text{ti}} \cup \mathcal{P}_{\text{te}}, \quad \mathcal{P}_a \cap \mathcal{P}_b = \emptyset \quad (a \neq b).$$

- ▶ Training updates weights and fits preprocessing; validation selects models and stopping.
- ▶ Interpolation cases lie inside the training convex hull; extrapolation cases lie outside.
- ▶ All realizations sharing one `physical_case_id` stay in one partition.
- ▶ The split manifest is frozen before model development.

All fitted transforms obey

$$\hat{\alpha} = \text{FitTransform}(\mathcal{D}_{\text{tr}}).$$

Refitting a transform on a held-out trajectory leaks its field values.

# Learning views do not overwrite truth

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The canonical release remains immutable. For  $(x, y, t, \eta, \nu, A_0) \mapsto (\psi, U)$ , periodic coordinates become

$$\mathbf{e}_{\text{per}} = (\sin \phi_x, \cos \phi_x, \sin \phi_y, \cos \phi_y), \quad \phi_x = 2\pi x/L_x, \quad \phi_y = 2\pi y/L_y.$$

Positive transport coefficients may use  $r_\eta = \log_{10} \eta$  and  $r_\nu = \log_{10} \nu$ . Outputs use training-only standardization:

$$\tilde{q} = \frac{q - m_q}{s_q}, \quad q = m_q + s_q \tilde{q}, \quad q \in \{\psi, U\}.$$

The view records chain-rule factors for PINN derivatives:

$$\partial_x q = \frac{s_q}{s_x} \partial_x \tilde{q}, \quad \partial_x^2 q = \frac{s_q}{s_x^2} \partial_x^2 \tilde{q}.$$



# Protecting the M3D-C1 claim

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Validation is layered:

structure  $\rightarrow$  relations  $\rightarrow$  numerics  $\rightarrow$  physics  $\rightarrow$  provenance  $\rightarrow$  release.

Readers verify axes, finite values, gauges, derived fields, diagnostics, checksums, disjoint splits, and parent links. Unknown major versions are rejected; migrations create traceable files.

M3D-C1-shaped = metadata + mesh/coordinates + parameters  
+ fields + diagnostics + splits.

This is an interface pattern, not physical equivalence. Genuine exports remain source-native; an adapter must document geometry, units, signs, gauges, projections, closures, and information loss before an M3D-C1 surrogate claim is justified.

# Module 4: A data-only field surrogate

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The parameterized coordinate network approximates a family of fields:

$$\mathcal{N}_{\theta} : (x, y, t, \eta, \nu, A_0) \mapsto (\hat{\psi}, \hat{U}).$$

Data are split by complete parameter cases:

$$\mu_c = (\eta_c, \nu_c, A_{0,c}).$$

- ▶ No point from a held-out case enters training or preprocessing.
- ▶ Parameter interpolation and extrapolation are separate; interpolation within  $t \in [0, T]$  is not forecasting beyond  $T$ .

# Periodic inputs respect the domain

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$$\phi_x = \frac{2\pi x}{L_x}, \quad \phi_y = \frac{2\pi y}{L_y},$$

$$\mathbf{z} = (\sin \phi_x, \cos \phi_x, \sin \phi_y, \cos \phi_y, \tilde{t}, \tilde{r}_\eta, \tilde{r}_\nu, \tilde{A}_0).$$

Smooth functions of  $\mathbf{z}$  are periodic in value and derivatives. Scaling statistics come from training cases only.

**Baseline:** six fully connected tanh layers of width 128 and a linear two-output head.  
Module 5 retains the same architecture family and periodic representation.

# Training uses labeled field error only

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For standardized output errors

$$e_{\psi,n} = \hat{\tilde{\psi}}_n - \tilde{\psi}_n, \quad e_{U,n} = \hat{\tilde{U}}_n - \tilde{U}_n,$$

the case-balanced objective is

$$\mathcal{L}_{\text{train}} = \mathcal{L}_{\text{data}} = \frac{1}{|\mathcal{C}_{\mathcal{B}}|} \sum_{c \in \mathcal{C}_{\mathcal{B}}} \left[ \frac{1}{N_c} \sum_{n \in c} e_{\psi,n}^2 + \frac{1}{N_c} \sum_{n \in c} e_{U,n}^2 \right].$$

There is no PDE, initial-condition, boundary-condition, gauge, global-balance,  $J$ ,  $\omega$ , derivative, or spectral target loss. Adam, mini-batches, regularization, and early stopping are optimization choices; they do not make the model physics-informed.

# The baseline makes the PINN testable

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Evaluation uses complete unseen fields, not training loss alone. For  $a \in \{\psi, U, J, \omega\}$ ,

$$\varepsilon_{L_2}^{(a)} = \frac{\|\hat{a} - a\|_2}{\|a\|_2 + \epsilon}.$$

$$\hat{J} = -\nabla^2 \hat{\psi}, \quad \hat{\omega} = \nabla^2 \hat{U},$$

Derived fields, energy, reconnection measures, periodicity, and PDE residuals are evaluation-only diagnostics.

**Controlled comparison:** Modules 4 and 5 share splits, labels, preprocessing, periodic encoding, architecture family, data loss, metrics, and seeds. Physics helps only if the matched PINN improves a declared metric, with added compute disclosed.

# Module 5: PDE residuals enter training

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The network still predicts  $(\hat{\psi}, \hat{U})$ , while automatic differentiation constructs

$$\hat{J} = -\nabla^2 \hat{\psi}, \quad \hat{\omega} = \nabla^2 \hat{U}.$$

The local reduced-MHD equation defects are

$$R_{\psi} = \hat{\psi}_t + [\hat{U}, \hat{\psi}] - \eta \nabla^2 \hat{\psi},$$

$$R_{\omega} = \hat{\omega}_t + [\hat{U}, \hat{\omega}] - [\hat{J}, \hat{\psi}] - \nu \nabla^2 \hat{\omega}.$$

At unlabeled collocation points, the PINN receives no solver field values; it is penalized when these residuals are nonzero.

# The objective combines five constraints

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$$\mathcal{L}_{\text{total}} = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{PDE}}\mathcal{L}_{\text{PDE}} + w_{\text{IC}}\mathcal{L}_{\text{IC}} + w_{\text{BC}}\mathcal{L}_{\text{BC}} + w_{\text{global}}\mathcal{L}_{\text{global}}.$$

- ▶  $\mathcal{L}_{\text{data}}$ : the identical labeled observations used by Module 4.
- ▶  $\mathcal{L}_{\text{PDE}}$ : normalized mean-square  $R_{\psi}$  and  $R_{\omega}$  at collocation points.
- ▶  $\mathcal{L}_{\text{IC}}$ : selects the intended trajectory at  $t = 0$ .
- ▶  $\mathcal{L}_{\text{BC}} = 0$  for hard periodicity;  $\mathcal{L}_{\text{global}}$  contains gauge, mean-flux, and optional balances.

# The vorticity residual reaches fourth order

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Because  $\hat{\omega} = \nabla^2 \hat{U}$ ,

$$\hat{\omega}_t \sim \partial_t \partial_x^2 \hat{U}, \quad [\hat{U}, \hat{\omega}] \sim \partial_x^3 \hat{U}, \quad \nabla^2 \hat{\omega} = \nabla^4 \hat{U}.$$

The Lorentz bracket also contains third derivatives of  $\hat{\psi}$ . Therefore smooth activations such as tanh are required, scaling factors must remain inside the differentiation graph, and derivative precision must be tested.

The PDE cannot identify a spatially constant offset in  $U$ , so a gauge condition remains necessary:

$$\langle \hat{U} \rangle(t, \mu) = 0.$$



# Balances and curricula stabilize training

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$$\frac{d}{dt} \frac{1}{2} \int_{\Omega} (|\nabla \hat{\psi}|^2 + |\nabla \hat{U}|^2) dA + \eta \int_{\Omega} \hat{J}^2 dA + \nu \int_{\Omega} \hat{\omega}^2 dA = 0.$$

Collocation defines the residual norm minimized. Uniform coverage may be supplemented by time-aware, structure-aware, or residual-adaptive points; validation points remain fixed.

Training proceeds in stages:

derivative tests  $\rightarrow$  data warm start  $\rightarrow R_{\psi} \rightarrow R_{\omega}$   
 $\rightarrow$  time curriculum  $\rightarrow$  parameterized cases.

Global penalties follow quadrature verification.

# Physics must add measurable value

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Module 6 compares matched models on unseen cases using field errors, residuals, condition defects, global balances, and cost.

- ▶ A PINN may improve labeled-data efficiency, derivative fidelity, parameter generalization, or physical admissibility.
- ▶ It may also train more slowly, oversmooth current sheets, suffer gradient conflict, or inherit error from an incomplete physical model.
- ▶ Reports include repeated seeds, failures, extrapolation, and data-versus-compute fairness.

**Scientific conclusion:** the PINN is not assumed to win. The result remains a surrogate of synthetic two-field reduced-MHD dynamics; an M3D-C1 surrogate requires genuine M3D-C1 data and Module 7.